



RoadAssist
B H A R A T

PROJECT REPORT · TOOLS AND SOFTWARE

What we use, what it does, and why we chose it

Every language, framework, cloud service, data source, developer tool and licence behind RoadAssist Bharat, an offline-first roadside-assistance and road-safety platform for India. Every version and number is taken from the repository.

SWE4004: Cloud Computing and Applications

VIT-AP University

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How to read this report. Each section starts in plain language and then gives the technical detail. Every tool, version and number comes from the repository: the package manifests and lock file, the Gradle and Python requirement files, the Dockerfile, `render.yaml`, the GitHub workflows, the project documents, and `app/docs/measured.json` for every count. The live setup is the one described in `DEPLOYMENT.md`. The same content is kept as `docs/TOOLS-AND-SOFTWARE.md` in the repository.

Status labels

- IMPLEMENTED** built and running
- SIMULATED** real software fed with made-up data, labelled so on screen
- MOCK** a stand-in for a paid service, so nothing real happens
- PREPARED** written and tested, but not switched on in the live demo
- FUTURE SCOPE** planned, not built

1 The system on one page

In plain language

RoadAssist Bharat helps a driver who has broken down on an Indian road. The driver describes the problem, the system works out the likely fault, finds the nearest verified mechanic, offers the job, and tracks the mechanic to the driver. If the driver is in danger, an SOS button raises an emergency. If there is no mobile data, the emergency is stored on the phone and sent automatically when the signal returns. A second part, **RAKSHA**, lets road authorities see road damage (such as potholes) that a trained camera model has found.

Everything runs as **one program on one server, with one database**. People reach it in five ways:

Way in	Who uses it	What it is	Status
Citizen web app	Drivers	A web page that installs like an app (a PWA) and keeps working offline. <code>/app.html</code>	IMPLEMENTED
Android app	Drivers	A native app written in Kotlin. It talks to the same server. <code>mobile/</code>	IMPLEMENTED
Mechanic console	Mechanics	A web page for accepting jobs and updating their status. <code>/mechanic.html</code>	IMPLEMENTED
RAKSHA authority dashboard	Road-safety officers	A web dashboard of detected road damage, with a live map. <code>/raksha.html</code> , <code>/map.html</code>	IMPLEMENTED (detections are real model output; their positions are SIMULATED)
SMS from a feature phone	Anyone with a basic phone	A complete booking by text message, in eight languages, with no app. <code>POST /v1/telecom/sms</code>	IMPLEMENTED in the API. The live demo has no SMS gateway connected , so outgoing messages are printed to the server log instead of sent

A sixth page, `/layers.html`, shows the architecture as a live 3D model that reads real figures from the running platform.

Not built, and the project says so: iOS, Android Auto, IVR, USSD, satellite, mesh networking, Kubernetes, Terraform, autoscaling and multi-zone hosting (**FUTURE SCOPE**, from the README's "Deliberately not built").

RoadAssist Bharat: how the pieces fit

One API and one database serve five ways in. The platform and its database run on free plans in Singapore.

solid line = live today
dashed = simulated, mock or not connected

1 • WHO USES IT

Citizen web app (PWA)
Plain HTML/CSS/JS, service worker, encrypted offline queue (IndexedDB)

Android app
Kotlin + Jetpack Compose
SOS ladder: data, SMS, 112, queue

Mechanic console
Browser page: accept jobs, update status, go on/off duty

RAKSHA dashboard + live map
Authority View: hazards, corridor conditions, review queue (Leaflet)

Feature phone (SMS only)
Book or raise SOS by text message, 8 languages, no app needed

RAKSHA edge device
YOLO11 detections (real model output); device + GPS positions SIMULATED

2 • EDGE

Cloudflare
Proxy in front of Render
HTTPS (TLS) for visitors
Real client IP via CF-Connecting-IP

SMS gateway
MSG91 / Twilio adapters written, NOT connected in the demo

3 • THE PLATFORM (RENDER)

Render web service
Docker image · free plan · Singapore
app: roadassistbharat.online
kept awake by keep-alive + UptimeRobot pings

One Node 22 process
Fastify 5 API: 74 routes, zod checks every input
JWT sessions (jose) + rotating refresh cookie
Serves every web page (no separate frontend)
Modules: auth · bookings · dispatch · emergency telecom (SMS) · payments · RAKSHA · geo
Live updates: server-sent events

Neon PostgreSQL '16 + PostGIS
Managed database, Singapore (ap-southeast-1), TLS
58 tables · 9 migrations (Drizzle ORM)
Hash-chained, append-only audit log

4 • OUTSIDE SERVICES (SERVER CALLS ONLY)

Open data (no keys)
OpenStreetMap map tiles (proxied, cached)
Nominatim: address for a point
Overpass: nearby hospitals, police, fuel
OSRM: road distance and driving time
Open-Meteo: weather and air quality
USGS: recent earthquakes
Positions rounded to ~110 m or ~1 km before they leave the server

Resend (email)
Alerts to the owner and sign-in codes, from a verified domain

Payments: Razorpay adapter
Written and stub-tested, but the demo runs PAYMENTS_PROVIDER=mock: no money ever moves

5 • HOW CODE REACHES THE INTERNET

GitHub repository
push to main

Render auto-deploy
builds app/Dockerfile itself

GitHub Actions
CI · publish-image · pages

GHCR image
for local demo + AWS kit

GitHub Pages
roadassistbharat.online

the same push also runs the workflows

publishes the showcase

The architecture in technical terms

- **A modular monolith** (ADR-0001): one Fastify process holds every module (auth, bookings, dispatch, emergency, telecom, payments, RAKSHA, geo) and also serves every web page. Module boundaries are enforced by a script in CI, not by convention (ADR-0002).
- **One database:** PostgreSQL 16 with the PostGIS extension (ADR-0003). The schema has 58 tables, 142 indexes and 9 migrations (`measured.json`).
- **74 routes**, 71 of them under `/v1` ; the other three are `/tiles` , `/basemap` and `/health` .
- **Real-time updates** use server-sent events with a heartbeat, and polling underneath (ADR-0010).
- **Off-Grid Mode is part of the design, not a fallback** (ADR-0009). The client tells apart `ONLINE` , `LIMITED` and `OFF-GRID` , keeps an encrypted on-device journal, and replays it without creating duplicates when the network returns.
- **A model is never allowed to dispatch** (ADR-0005, ADR-0011). An automatic signal waits in `AWAITING_CONFIRMATION` until a person or a second signal confirms it.

The project records its design choices in 13 ADRs under `app/docs/adr/` .

2 Languages and frameworks

In plain language

The server is written in **TypeScript** (JavaScript with types) and runs on **Node.js**. The web pages are written in plain **HTML**, **CSS** and **JavaScript**, with no build step. The Android app is written in **Kotlin**. The road-damage detector is trained and run with **Python**. Data is stored in **PostgreSQL**, a database, with **PostGIS**, which adds maps and distances to it.

Versions below are the ones the project actually installs: npm packages from `app/package-lock.json`, Android libraries from `mobile/app/build.gradle.kts` (the project has no `gradle/libs.versions.toml`; versions are written inline), and Python packages from the pinned `ai/requirements*.txt` files.

2.1 Server (the API)

Tool	Version	What it does here	Why it was chosen	Where
Node.js	22 (<code>"node": ">=22"</code> ; image <code>node:22-alpine</code> ; CI <code>node-version: 22</code>)	Runs the API	One language across server and browser. Node 22 is needed because the browser test suite uses the built-in <code>WebSocket</code>	<code>app/package.json</code> , <code>app/Dockerfile</code>
TypeScript	5.9.3	Adds types to the server code; checked with <code>tsc -b</code>	Catches mistakes before the code runs	<code>app/tsconfig.json</code>
Fastify	5.12.5	The web framework: routes, hooks, logging	Fast, with a plugin model that suits a modular monolith	<code>app/apps/api/src/server.ts</code>
@fastify/static	10.1.5	Serves the web pages and demo media from the same process	No separate web host, so no second deploy and no CORS boundary	<code>server.ts</code>
@fastify/cors	10.1.0	Allows only the listed origins (the showcase and the platform)	An unset value would reflect any origin	<code>render.yaml</code> <code>CORS_ORIGINS</code>
zod	3.25.76	Validates the input of every route	One schema both checks and types the data	<code>apps/api/src/</code>
jose	5.10.0	Signs and verifies JWT access tokens	Standards-based; refuses <code>alg:none</code> and forged signatures	<code>apps/api/src/auth.ts</code>
Drizzle ORM	0.45.3	Typed database queries and the schema definition	SQL stays visible; schema lives in TypeScript	<code>app/packages/db/</code>
drizzle-kit	0.31.11 (dev only)	Generates migration files from the schema	Pairs with Drizzle ORM; never shipped in the image	<code>packages/db/package.json</code>
postgres (postgres.js)	3.4.9	The PostgreSQL driver	Small, fast, no native build step	<code>packages/db/package.json</code>

Tool	Version	What it does here	Why it was chosen	Where
@faker-js/faker	10.6.0 (dev only)	Generates demo users, mechanics and bookings for seeding	Realistic test data; kept out of the production image on purpose	packages/db/src/seed.ts
tsx	4.23.15 (dev only)	Runs TypeScript directly in development and tests	No compile step while developing	npm run dev, npm test

The production image runs compiled JavaScript (`tsc` output in `dist/`) with production dependencies only; the compiler, linter and drizzle-kit are left out because each is attack surface (`app/Dockerfile` stage 3).

2.2 Database

Tool	Version	What it does here	Why
PostgreSQL	16	Stores everything: users, bookings, incidents, payments, RAKSHA detections, audit log	Transactions and row locks settle races (for example two mechanics accepting one job)
PostGIS	3.4 locally and in CI (image <code>postgis/postgis:16-3.4</code>); the PostGIS extension on Neon	Geometry columns, GiST spatial indexes, and <code>ST_DWithin</code> to rank the nearest mechanics	A hard requirement: the first migration runs <code>CREATE EXTENSION postgis</code> , which ruled out free hosts without it (DEPLOYMENT.md)

Database facts from `measured.json` : 58 tables, 63 foreign keys, 142 indexes (88 unique, 5 GiST), 9 migrations. The audit log is append-only: Postgres `RULES` turn `UPDATE` and `DELETE` into no-ops, and each row is hash-chained to the one before it.

2.3 Web surfaces (citizen app, mechanic console, RAKSHA, map, 3D layers)

Tool	Version	What it does here	Where
Plain HTML, CSS, JavaScript	n/a	Every page. No framework and no bundler	app/apps/web/
Service worker	cache <code>ra-v19</code>	Keeps the app shell and map tiles available offline. API answers are never cached, because a stale booking status is worse than an honest failure (ADR-0004)	app/apps/web/sw.js
IndexedDB + Web Crypto (AES-GCM-256)	Browser built-ins	The offline SOS and sync journal. Payloads are encrypted under a non-extractable key generated on the device	app/apps/web/offline-store.js , offline-engine.js
Web app manifest	n/a	Lets the citizen app install to the home screen (<code>start_url</code> /app.html , standalone)	app/apps/web/manifest.webmanifest
Leaflet	1.9.4 (vendored)	The maps on the live map and the RAKSHA dashboard	vendor/leaflet.js , used by map.html and raksha.html

Tool	Version	What it does here	Where
Leaflet.markercluster	vendored; the file carries no version string	Groups nearby markers on the live map	vendor/markercluster.js , map.html
three.js	r169 (vendored, MIT)	Draws the 3D architecture model: OrbitControls, GLTF and Draco loaders, bloom post-processing	vendor/three/ , layers.html
Server-sent events (EventSource)	Browser built-in	Live booking and dispatch updates	apps/api/src/realtime.ts

Why no framework and no CDN: the pages are served by the API itself, so there is one deployment, and every library and font is copied into `vendor/` so that no page makes a visitor's browser contact a third party (enforced by `npm run residency`, see section 6).

2.4 Android app

Tool	Version	What it does here
Kotlin	2.0.21 (<code>org.jetbrains.kotlin.android</code> , Compose compiler plugin 2.0.21)	The app's language
Android Gradle Plugin	8.13.2	Builds the app
Gradle (wrapper)	8.13	The build tool. Must run on JDK 21 : Gradle 8.13 rejects JDK 25 (ENGINEERING-NOTES.md)
Java target	17 (<code>sourceCompatibility</code> , <code>jvmTarget</code>)	Bytecode level of the app
Android SDK	<code>compileSdk</code> 35, <code>targetSdk</code> 35, <code>minSdk</code> 26	Runs on Android 8.0 and newer
Jetpack Compose	BOM 2024.09.03; <code>ui</code> , <code>foundation</code> , <code>material3</code>	The user interface
<code>androidx.activity:activity-compose</code>	1.9.2	Hosts Compose in the activity
<code>androidx.lifecycle:lifecycle-runtime-ktx</code>	2.8.6	Lifecycle-aware coroutines
<code>androidx.core:core-ktx</code>	1.13.1	Status-bar and navigation-bar colours for the light/dark toggle
<code>kotlinx-coroutines-android</code>	1.8.1	Background work (network, SOS ladder)
<code>androidx.profileinstaller</code>	1.4.1	Installs a baseline profile so start-up and scrolling are compiled ahead of time
R8 (release build)	via AGP	Shrinks and optimises the release APK. The build notes record about 25% janky frames on a debug build against about 4% on the R8 build
JUnit	4.13.2 (tests only)	Unit tests
<code>org.json</code>	20240303 (tests only)	A real JSON library for JVM tests; the Android SDK's copy is a stub off-device

Why so few libraries: networking uses the platform's own `HttpURLConnection` and `org.json`, so the app has no third-party runtime library at all ("zero third-party libraries, so the first build has the fewest possible failure modes", `mobile/app/build.gradle.kts`). The release APK is signed with the debug key so it installs on a demo device; it is **not** a store distribution build.

The SOS ladder's decisions (data, then SMS, then 112, then the on-device queue) live in `SosLadder.kt` as pure functions so they can be tested off-device.

2.5 AI: RAKSHA road-damage detection (Python)

Tool	Version	What it does here
Python	3.12 (<code>ai/requirements.txt</code> , CI <code>python-version: '3.12'</code>)	Training, inference and the pipeline tests
Ultralytics (YOLO11)	8.4.126	Trains and runs the YOLO11 detector. Pulls in PyTorch, about 2 GB installed. AGPL-3.0 (see section 7)
remotezip	0.12.5	Downloads only the needed byte ranges of the 13.26 GB RDD2022 archive
ONNX Runtime	1.30.0	Runs the exported <code>best.onnx</code> model in the serving path, without PyTorch or Ultralytics
NumPy	2.5.3	Pre- and post-processing for ONNX inference
Pillow	12.3.0	Image loading for inference
<code>unittest</code> (standard library)	n/a	The pipeline tests: class mapping, box maths, severity rule. No PyTorch needed, so they run on every push

What was trained (from `ai/README.md`, measured on held-out validation data, CPU-only training):

Model	Classes	mAP50	mAP50-95	ONNX size
<code>yolo11n</code> baseline (India)	2	0.443	0.183	10 MB
<code>yolo11s-multi-rich</code> (India, Czech, Japan, USA)	4	0.472	0.226	37 MB
<code>yolo11n-multi-edge</code>	2	0.293	0.117	10 MB

The detections the live RAKSHA dashboard shows are real output of the YOLO11n model (`yolo-rdd2022in-best`) on RDD2022 India images. Their positions on NH-48 are **SIMULATED**, because RDD2022 images carry no GPS. On the hosted demo, the API registers a demo patrol device at start-up and posts the 34 detections from `ai/cv-detections-full.json` through the real ingest route. Locally, `npm run demo:raksha` posts 65 detections from `ai/cv-live-show.json`.

`ai/serve.py` puts the ONNX model behind `POST /detect` over HTTP. It runs on a developer machine; it is not deployed to the cloud (**IMPLEMENTED** locally). A physical edge device (the README names "Pi-class hardware" as the target) is **FUTURE SCOPE**.

The roadside fault diagnosis is not AI. It is a deterministic rule table that returns `rules-1.0.0`, and every screen that shows it says it is a rules engine (ADR-0006). No language model is used anywhere; `npm run no-llm` fails the build if one appears. The claim "AI-powered" is recorded as **PARTIAL** in CLAIMS-AUDIT.

2.6 What was planned but not used

The original setup guide (`setup-guide/tools-and-setup.md`) planned pnpm, Node 20, Kubernetes (k3d), Terraform, k6, OWASP ZAP, Grafana, Sentry and Postman. The project as built uses **npm workspaces** and **Node 22** instead, and does not use the others. Telemetry SDKs such as Sentry are refused outright by the data-residency gate, because they send identifiers abroad by default.

`app/docker-compose.yml` also starts **Redis 7** and **Redpanda v24.2.7** (a Kafka-compatible event bus) as local stand-ins for managed services. The API does not use either today: rate limits are kept in memory, and `ratelimit.ts` names Redis as the fix for sharing them across several servers (`FUTURE SCOPE`).

3

Cloud and hosting

In plain language

The platform lives on **Render**, a hosting company that runs our program in a **Docker container** (a sealed box holding the program and everything it needs). The data lives on **Neon**, a hosted PostgreSQL database. Both are in **Singapore**, the closest region to India that their free plans offer. **Cloudflare** sits in front and handles secure (HTTPS) connections. A separate public showcase page is hosted free on **GitHub Pages**. **GitHub Actions** tests every change automatically.

Addresses

Address	What it is	Hosted on
https://app.roadassistbharat.online	The live platform: API and every web surface	Render, behind Cloudflare
https://roadassistbharat.online	The showcase page, which embeds the live platform	GitHub Pages (<code>pages/CNAME</code>)

Services

Service	What it does here	Configuration	Status
Render	Runs the Docker image as one web service. Rebuilds and redeploys on every commit (<code>autoDeployTrigger: commit</code>). Health check <code>/health</code> , which answers 503 when the database is unreachable	<code>render.yaml</code> : <code>runtime: docker</code> , <code>plan: free</code> , <code>region: singapore</code> , <code>NODE_ENV=demo</code>	IMPLEMENTED (live)
Neon	Managed PostgreSQL 16 + PostGIS, Singapore (<code>ap-southeast-1</code>), TLS required. The direct (not pooled) connection string is used, because migrations and live streams need a real session	Set by hand as <code>DATABASE_URL</code> in <code>Render</code> ; never committed	IMPLEMENTED (live)
Least-privilege database role	<code>roadassist_app</code> can read and write rows only: no <code>CREATE</code> , <code>TRUNCATE</code> or ownership, and only <code>SELECT / INSERT</code> on <code>audit_log</code> . Migrations run as the owner through <code>MIGRATION_DATABASE_URL</code> , which <code>docker-start.sh</code> uses for the migrate step and then removes	<code>app/packages/db/sql/least-privilege-role.sql</code> , <code>DEPLOYMENT.md</code>	PREPARED and verified locally. <code>app/docs/SECURITY.md</code> (30 Sep 2026) and the showcase list applying it on the hosted database as pending, while <code>deploy/aws/env.example</code> (4 Oct 2026) describes <code>Render's DATABASE_URL</code> as the <code>roadassist_app</code> string. The repository does not settle which is current; <code>Render's Environment</code> page does
Cloudflare	Proxy in front of <code>Render</code> . Visitors' HTTPS ends here, and the API takes the caller's address from <code>CF-Connecting-IP</code> , which a caller cannot forge	<code>CLIENT_IP_HEADER=cf-connecting-ip</code> , <code>TRUST_PROXY=true</code> (<code>render.yaml</code>)	IMPLEMENTED (live)

Service	What it does here	Configuration	Status
Hostinger	Where the domain is managed. DEPLOYMENT.md §3 adds the <code>app</code> CNAME and the Resend DKIM/SPF records there; the newer AWS kit edits the <code>app</code> record in Cloudflare's DNS	DNS records	IMPLEMENTED
GitHub Pages	Hosts the static showcase. Built and checked by <code>pages.yml</code> , which fails if the page loads anything from another origin	<code>.github/workflows/pages.yml</code> , <code>pages/</code>	IMPLEMENTED (live)
GitHub Container Registry (GHCR)	Stores the published image <code>ghcr.io/saatwik-1157/roadassist-bharat</code> , tagged by branch, version and commit SHA so a bad deploy can roll back to a known image	<code>.github/workflows/publish-image.yml</code>	IMPLEMENTED
GitHub Actions	Runs the checks and publishing (table below)	<code>.github/workflows/*.yml</code>	IMPLEMENTED
UptimeRobot	An uptime monitor that pings the platform. Its settings live in the UptimeRobot dashboard, not in the repository; <code>deploy/aws/README.md</code> is the only file that names it	UptimeRobot dashboard	IMPLEMENTED (configured outside the repo)
Resend	Sends operator alert emails and email sign-in codes from the verified domain <code>send.roadassistbharat.online</code>	<code>EMAIL_PROVIDER=http</code> , <code>EMAIL_BASE_URL=https://api.resend.com/emails</code>	IMPLEMENTED (live)
Google Search Console	Both sites are verified URL-prefix properties and both sitemaps are submitted (done 2026-09-30). The verification files <code>google923ee0decf8e5e0.html</code> must not be deleted	DEPLOYMENT.md, "Search engines"	IMPLEMENTED
Fly.io	A ready alternative to Render that runs the same image, region Mumbai (<code>bom</code>)	<code>fly.toml</code>	PREPARED, not live
AWS (free plan)	One EC2 <code>t3.micro</code> (Amazon Linux 2023, Singapore) running the same GHCR image with Caddy 2 as the HTTPS origin behind Cloudflare (SSL mode "Full"), against the same Neon database. Switching is one DNS edit	<code>deploy/aws/README.md</code> , <code>user-data.sh</code> , <code>env.example</code>	PREPARED, not live

GitHub Actions workflows

Workflow	Runs on	What it does
<code>ci.yml</code>	Every push to <code>main</code> and every pull request	Five jobs. verify : install, typecheck, lint, unit tests, <code>gitLeaks</code> secret scan, <code>npm audit --audit-level=high</code> . integration : a real PostGIS service, migrate, seed, start the API, then the end-to-end, gateway-security, concurrency, security-audit and browser suites; a chaos test that stops PostgreSQL under the running API; a backup-and-restore rehearsal that re-checks the audit log; and an offer-expiry run with a 2-second offer lifetime. boundaries : module boundaries, claims, citations, no-LLM and data-residency gates. android : lint, unit tests, debug APK and release APK on JDK 21 (Temurin). ai : syntax check, the Python pipeline tests, and a check that every requirement is pinned with <code>==</code>
<code>publish-image.yml</code>	Push to <code>main</code> , <code>v*</code> tags, manual	Builds the image, boots it against a real PostGIS, runs the end-to-end and security suites against the running container , checks every surface answers 200 and that <code>/media</code> will not serve an HTML page, then pushes to GHCR
<code>pages.yml</code>	Changes under <code>pages/</code> , screenshots, videos, deck visuals	Assembles the showcase, refuses any third-party subresource, checks every referenced asset exists, deploys to GitHub Pages
<code>keep-awake.yml</code>	Every 10 minutes (cron), manual	A plain GET of <code>/health</code> and the showcase, so the free Render service does not sleep. Sends no credentials

Actions used: `actions/checkout@v5`, `actions/setup-node@v5`, `actions/setup-java@v4`, `actions/setup-python@v5`, `android-actions/setup-android@v3`, `gradle/actions/setup-gradle@v4`, `gitLeaks/gitLeaks-action@v2`, `docker/setup-buildx-action@v3`, `docker/build-push-action@v6`, `docker/login-action@v3`, `docker/metadata-action@v5`, `actions/upload-artifact@v4`, `actions/upload-pages-artifact@v3`, `actions/deploy-pages@v4`.

The Docker image

`app/Dockerfile`, built from the repository root:

- deps**: `node:22-alpine`, `npm ci` from the manifests only, so this layer is cached until a dependency changes.
- build**: compiles `packages/db` and `apps/api` with TypeScript.
- prod-deps**: `npm ci --omit=dev`, so no compiler, linter or drizzle-kit ships.
- runtime**: runs as the unprivileged `node` user, uses `tini` as PID 1 so shutdown signals reach Node, and has a `HEALTHCHECK` on `/health`.

On start, `docker-start.sh` runs the idempotent migrations, optionally seeds the demo fleet (24 mechanics marked "(simulated)") and the RAKSHA admin and corridor, then starts the server.

Why these hosts

- PostGIS was the deciding requirement.** A managed Postgres that will not install the extension fails on the first migration.
- The database is on Neon, not Render**, because Render allows one free PostgreSQL per account, and this account's was already in use.
- Netlify, Vercel and other function-per-request hosts cannot run this API.** It holds open live-update streams, runs an offer-expiry timer and keeps a database pool, and all three need a long-lived process

(DEPLOYMENT.md).

- **Singapore, not India:** the free tiers offer no Indian region. The project's rule "no personal data leaves India" is therefore the production target and not yet true of the demo, and the README says so.
- `NODE_ENV=demo`, **not** `production`: the production guard refuses to start without a real payment gateway, a real SMS provider and webhook secrets. Running as `demo` keeps that guard meaningful, and the screens say payments are mocked.

4 Third-party data services

In plain language

The app shows nearby hospitals and fuel stations, a readable address, road driving time, weather and air quality, and recent earthquakes. That information comes from free public services. **Only our server talks to them, never the visitor's browser**, and a user's position is rounded before it is sent so the service learns only a rough area.

Declared outbound hosts

`app/scripts/check-data-residency.mjs` lists every host the project's own code may contact, with where the data lands. An undeclared host fails the build. The declarations, summarised:

Host	Used for	What it receives	Region (as declared)	Status in the live demo
<code>tile.openstreetmap.org</code>	Map tiles, fetched by the server and cached (<code>/tiles</code> , <code>/basemap</code>)	Tile coordinates only	Outside India (EU)	IMPLEMENTED
<code>nominatim.openstreetmap.org</code>	Address for a point (<code>GET /v1/geo/address</code>)	Position rounded to 3 decimals (about 110 m)	Outside India (OSMF, EU)	IMPLEMENTED
<code>overpass-api.de</code>	Nearby hospitals, police, fuel, EV charging, repair, tyres (<code>GET /v1/geo/nearby</code>)	Point rounded to 2 decimals (about 1 km) and a radius	Outside India (Germany)	IMPLEMENTED
<code>router.project-osrm.org</code>	Road distance and driving time, no live traffic (<code>GET /v1/geo/route</code>)	Both points rounded to about 110 m	Outside India (FOSSGIS, Germany)	IMPLEMENTED
<code>api.open-meteo.com</code>	Weather for the RAKSHA corridor report	Coordinates (declared as personal data)	Outside India (Germany)	IMPLEMENTED
<code>air-quality-api.open-meteo.com</code>	Air quality in the corridor report; a "poor air quality" risk above US AQI 150	Fixed corridor midpoints, never a user's position	Outside India (Germany)	IMPLEMENTED
<code>earthquake.usgs.gov</code>	Magnitude 4+ earthquakes in and around India, last 7 days (<code>GET /v1/geo/earthquakes</code>)	A fixed area and a date, nothing about any user	Outside India (USA)	IMPLEMENTED
<code>api.resend.com</code>	Operator alerts and email sign-in codes	A phone number masked to its last three digits, a role, an event and a time	Outside India (USA)	IMPLEMENTED , needs <code>EMAIL_API_KEY</code>
<code>api.msg91.com</code>	SMS (the default provider in the code)	Phone number and message	India	PREPARED : adapter written; the demo runs <code>SMS_PROVIDER=console</code> , so messages go to the server log

Host	Used for	What it receives	Region (as declared)	Status in the live demo
api.twilio.com	Optional fallback SMS provider	Phone number and message	Outside India (USA)	PREPARED : adapter written, never used in a documented run
api.razorpay.com	Payment orders	Order and payment references, never card data	India	PREPARED : adapter written and checked against a local stub; the demo runs PAYMENTS_PROVIDER=mock , so no money moves
ndownloader.fileshare.com	Downloading the RDD2022 dataset at training time	Nothing; a download only	Outside India	Training only, never called by the platform

The only third-party host a **page** may load is `checkout.razorpay.com` , because card data must load from the gateway's own origin. It is not used in the demo, where payments are mocked.

How the location calls are kept polite and safe (README, ADR-0012): results are cached (address 24 h, nearby 6 h, route 10 min, earthquakes 15 min, air quality 30 min), calls are paced to about one a second, each carries an identifying User-Agent, and a point outside India is refused with `400 outside_region` before any provider is asked. They are off in development, test and CI, so no test run calls a donated service. **GEO_SERVICES=on|off** overrides this.

Honest limits: rounding reduces what leaves but does not anonymise it, so sending rounded positions abroad is a stated exception to "no personal data leaves India". The production answer is a self-hosted Nominatim and OSRM in an Indian region (**FUTURE SCOPE**). Community map data can be missing or stale, and "nearby" distances are straight-line.

Simulated and mocked parts, said plainly

What	How it is simulated
RAKSHA detections	Real YOLO11 output on real RDD2022 images; the device and the NH-48 positions are SIMULATED
Mechanics, police and ambulance on the live map	Seeded and marked "(simulated)"; the 24 demo mechanics belong to "Demo Fleet (simulated)"
Payments	MOCK provider: logs "SIMULATED, no money moved"
SMS	No gateway: messages are printed to the server log, with numbers masked and codes redacted
Phone sign-in codes	Shown on screen (EXPOSE_DEV_OTP=true), and only for the published demo numbers; admin, RAKSHA officer and listed mechanic accounts sign in by emailed code only
ERSS-112 handoff	Stubbed, and the API response says so

5 Developer tools

In plain language

These are the tools the team uses to build, test and present the project. None of them ships to users.

Tool	What it does here	Where it shows up
Git and GitHub	Version control and the shared repository. The repository is pushed from one account, so <code>git log</code> shows one committer; that reflects how code reached GitHub, not how work was split (README)	Whole repo
npm workspaces	One install for the <code>@roadassist/api</code> and <code>@roadassist/db</code> packages (<code>apps/*</code> , <code>packages/*</code>). Run npm commands from <code>app/</code>	<code>app/package.json</code> <code>workspaces</code>
Docker and Docker Compose	<code>docker compose up -d db</code> starts a local PostGIS on port 5434. <code>docker compose -f docker-compose.demo.yml up</code> brings up the whole platform from the published image with one command. The setup guide installs Docker Desktop on the team's Windows machines	<code>app/docker-compose.yml</code> , <code>docker-compose.demo.yml</code> , <code>app/docker-compose.prod.yml</code>
tsx	Runs the API from TypeScript source in development (<code>npm run dev</code> watches files)	<code>apps/api/package.json</code>
TypeScript compiler	<code>npm run typecheck</code> (<code>tsc -b</code> , project references) and the production build	<code>app/tsconfig.json</code>
ESLint 9 + typescript-eslint 8	<code>npm run lint</code> ; ESLint 9.39.5 and typescript-eslint 8.71.0 with a flat config	<code>app/eslint.config.js</code>
node:test	The unit test runner built into Node (<code>npm test</code>)	<code>apps/api/test/</code> , <code>packages/db/test/</code>
drizzle-kit	<code>npm run db:generate</code> makes a migration; <code>drizzle-kit studio</code> browses the database	<code>packages/db/package.json</code>
Google Chrome	Driven over the DevTools Protocol by the browser test suite and the screenshot script, with no headless-browser library added. Headless Chrome also renders the showcase hero illustration and <code>ppt/md2pdf.py</code> 's PDFs	<code>scripts/ui-journey.mjs</code> , <code>scripts/capture-screens.mjs</code> , <code>pages/hero/build-hero.py</code>
cloudflared (Cloudflare quick tunnel)	<code>npm run share</code> gives a local run a temporary public HTTPS address, so a real phone gets geolocation, the service worker and "Add to home screen"	<code>scripts/share.mjs</code>
pg_dump / pg_restore	Backups, and the backup-and-restore rehearsal in CI	<code>scripts/backup.sh</code> , <code>ci.yml</code>
gitleaks	Scans every push for committed secrets	<code>ci.yml</code>
Android Studio + emulator	Builds and runs the Android app. From the emulator, a local API is at <code>10.0.2.2:4000</code> . Studio's Gradle JDK must be set to JDK 21 by name (ENGINEERING-NOTES.md)	<code>mobile/</code>
Gradle wrapper	<code>cd mobile && ./gradlew lint testDebugUnitTest assembleRelease</code>	<code>mobile/gradlew</code>
Python 3.12 + venv	The AI pipeline lives in <code>ai/.venv</code> (gitignored)	<code>ai/</code>

Tool	What it does here	Where it shows up
python-pptx + Pillow	Generate the final deck: 38 slides from <code>ppt/part_final.py</code> via <code>python ppt/make_final.py</code> . Never edit the <code>.pptx</code> by hand	<code>ppt/</code>
Microsoft PowerPoint (COM)	Applies the slide transitions and exports the deck PDF after each build	<code>ppt/README.md</code>
pptxgenjs	Builds a separate review deck from <code>measured.json</code> . It is not declared in any <code>package.json</code>	<code>design/deck/build.js</code>
Pillow scripts	Web-sized thumbnails, the 3D journey scenes and the deck visuals, all built from real screenshots	<code>pages/build-thumbs.py</code> , <code>pages/scenes/</code> , <code>ppt/render_visuals.py</code>
Blender 5.2	A headless script models the seven-layer architecture stack (7 layers, 31 parts), exports it as glTF with Draco compression (595 KB, <code>assets/3d/layers.glb</code>) for <code>three.js</code> , and renders the still shown before the live scene loads	<code>design/blender/build_layers.py</code>
Figma	The design boards are written as SVG that Figma imports as editable layers	<code>design/build_boards.py</code> , <code>design/figma/</code>
Promo film	A 30-second promo and a vertical cut, built from real recordings and screenshots, with drawn scenes tagged. The repository does not record which video tool assembled it	<code>site/roadassist-promo.mp4</code> , <code>site/roadassist-promo-vertical.mp4</code>

Everyday commands (from `app/`):

```

docker compose up -d db      # PostGIS on 5434
npm run demo:reset          # reset, migrate, seed
npm start                   # API on :4000, serves every web surface
npm run demo:raksha         # second terminal: real YOLO11 detections, simulated positions
npm run verify              # the eight gates listed in section 6

```

6 Quality and security

In plain language

Every change is checked by machines before it can go live. Some checks test that the software works; others check that the project's documents tell the truth, that no personal data is sent to outside services by a web page, and that no AI chat service has crept in. The security tests try more than a hundred attacks against the running system, and each one has to fail.

The verify gates

`npm run verify` runs these in order, and CI runs the same checks:

Gate	Command	What it fails on
Typecheck	<code>npm run typecheck</code>	Any TypeScript type error
Lint	<code>npm run lint</code>	ESLint errors
Boundaries	<code>npm run boundaries</code>	A module importing another module's internals; an incomplete offline shell; the app shell loading in the wrong order
Claims	<code>npm run claims</code>	Any document, deck source or page whose numbers disagree with <code>app/docs/measured.json</code>
Citations	<code>npm run citations</code>	A <code>file.ts</code> line reference in the viva packs that no longer points at real code
No LLM	<code>npm run no-llm</code>	A language-model SDK in any manifest, or a chat endpoint called from any source file
Residency	<code>npm run residency</code>	A page loading a third-party resource; an undeclared server-side host; an analytics or crash-reporting SDK; personal data in a URL
Unit	<code>npm test</code>	Any failing unit test

Test suites

Counts are from `app/docs/measured.json` (measured on 2026-10-01 against a database migrated from empty, then seeded).

Suite	Count	What it proves	Runs
Unit (<code>npm test</code>)	451	Pure logic: diagnosis rules, booking and incident state machines, backoff, log redaction, and a guard that the on-device rule table matches the server's	CI and <code>verify</code>
End-to-end (<code>npm run test:e2e</code>)	246	The whole API journey against real Postgres, including the SMS feature-phone journey and off-grid sync	CI, and against the built container
Concurrency (<code>npm run test:concurrency</code>)	92	Races: two mechanics accepting one job, three SOS taps at once, live event delivery	CI
Gateway security (<code>npm run test:gateway</code>)	58	Webhook signatures, append-only audit rules, sign-in code limits per number and per IP	CI

Suite	Count	What it proves	Runs
Security audit (<code>npm run test:security</code>)	106	Attacks that must all be refused: cross-tenant access, role escalation, SQL injection, forged and <code>alg:none</code> tokens, unsigned webhooks, oversized input, error leakage	CI, and against the built container
Browser (<code>npm run test:ui</code>)	184	Drives real Chrome: offline payment refused, session survives reload, the full Off-Grid Mode scenario	CI
Total	1137	Six suites, zero failures	
Payment sandbox (<code>npm run test:razorpay</code>)	22	Razorpay negative cases against a local stub. Needs an API started with <code>PAYMENTS_PROVIDER=razorpay</code> , so it is outside every npm test run, not part of the 1137, and never described as passing	By hand only
Android (Gradle)	182	Android unit tests, run with lint and both APK builds	CI <code>android</code> job
SOS ladder	21	The subset of the Android tests over <code>SosLadder.kt</code> , the emergency fallback decisions	CI <code>android</code> job
CV pipeline	39	The Python pipeline tests (standard library only)	CI <code>ai</code> job

CI also stops PostgreSQL under a running API and checks that `/health` answers 503 naming the database while `/v1/ping` still answers 200, so a client can tell "you have no network" from "the platform is unwell".

Security controls in place

Area	Control
Sign-in	One-time codes over SMS (or email for protected accounts), stored as SHA-256 hashes; limits per number and per IP counted in Postgres. The platform has no passwords
Sessions	10-minute JWT access tokens; rotating refresh tokens, and reusing an old one revokes the whole family
Refresh cookie	Web pages get the refresh token as an HttpOnly cookie on <code>Path=/v1/auth</code> (<code>SameSite=None; Secure; Partitioned</code> over HTTPS). Android and the test scripts use the JSON body token
Content Security Policy	<code>default-src 'self'</code> ; scripts only from the site itself plus each inline <code><script></code> allowed by its SHA-256 hash , computed when the server starts (<code>'wasm-unsafe-eval'</code> is allowed for the 3D page's Draco decoder, and <code>checkout.razorpay.com</code> only when Razorpay is configured); <code>object-src 'none'</code> ; <code>frame-ancestors</code> limited to the showcase. Also HSTS, nosniff, Referrer-Policy, Permissions-Policy, COOP/CORP and <code>no-store</code> on the API (<code>security-headers.ts</code>)
Rate limits	Per principal; the SMS webhook per sending number; map tiles bounded. SOS alerts are capped, but the escalation itself is never limited
Input	zod on every route; pages escape server strings; CSV exports neutralise formulas
Audit log	Append-only (Postgres <code>RULES</code>) and hash-chained; re-verified live and on every restored backup
Database role	Least-privilege <code>roadassist_app</code> role prepared and tested (see section 3 for its hosted status)
Location data	Positions rounded to about 110 m or 1 km before any outside call; logs never record coordinates, only whether a fix was available (<code>observability.ts</code>)
Secrets	None in source, image or history; <code>gitLeaks</code> on every push; the JWT secret must be 32+ characters outside development
Dependencies	<code>npm audit --audit-level=high</code> in CI. Four moderate advisories are knowingly carried, all in drizzle-kit's development-only <code>esbuild</code> chain, which never ships
Container	Runs as a non-root user; production dependencies only

What is not done (from `app/docs/SECURITY.md`)

- **No external penetration test.** 106 self-written attacks is not the same thing.
- **Rate limiting is per instance.** Several servers would each allow the full limit.
- **No column-level encryption at rest** in Postgres; medical data is protected by access control and audit.
- **No live payment gateway** has ever been exercised.
- **No WAF and no DDoS protection** beyond what the hosts provide.
- **Coarsened positions still leave India** wherever location services are on.
- The 10-minute access token sits in `sessionStorage`; `style-src` still allows `'unsafe-inline'`.
- No load test: latency is measured for a single user on one machine (README).

7 Fonts, assets and licences

In plain language

Every font, map library and 3D library is free to use under an open licence, and is copied into the project rather than loaded from someone else's server. The road-damage training images come from a public research dataset that must be credited. The detector software has a strict licence that the project addresses by keeping it out of the part that would ship.

Item	Licence	How the project uses it	Where
Space Grotesk	SIL Open Font License 1.1	Display headings	<code>app/apps/web/vendor/fonts/</code> , <code>pages/fonts/</code> , <code>mobile/app/src/main/res/font/</code>
Inter	SIL Open Font License 1.1	Everything a person reads or taps	same
JetBrains Mono	SIL Open Font License 1.1	Figures, references and codes (RA-XXXXXX, 112) where digits must line up	same
Font licence texts	n/a	<code>OFL-*.txt</code> shipped beside the fonts; the Android app carries them in <code>assets/licenses/</code>	
Leaflet 1.9.4	The vendored file keeps its copyright header but no licence file is vendored beside it; Leaflet's authors publish it under BSD-2-Clause	Maps; every map credits "© OpenStreetMap contributors"	<code>vendor/leaflet.js</code>
OpenStreetMap data	ODbL; the geo routes return the credit "© OpenStreetMap contributors, ODbL"	Tiles, addresses, nearby places	<code>apps/api/src/routes/geo.ts</code> , the map pages
three.js r169	MIT (<code>vendor/three/LICENSE</code>)	The 3D layers page	<code>vendor/three/</code>
RDD2022 dataset (Arya et al.)	figshare records CC BY 4.0; the maintainers' README states CC BY-SA 4.0 for images. The project follows the stricter reading	Training data; the dashboard frames are derivatives under the same licence, credited in the RAKSHA photo viewer, and the showcase links the paper (arXiv 2209.08538)	<code>docs/raksha/05-dataset-license-verification.md</code>
Ultralytics YOLO11	AGPL-3.0	Training and detection	<code>ai/requirements.txt</code>
ONNX Runtime, NumPy, Pillow	MIT, BSD-3, MIT-CMU (as recorded in <code>ai/requirements-serve.txt</code>)	The serving path	<code>ai/requirements-serve.txt</code>
npm packages	MIT for Fastify, zod, jose, ESLint, tsx and others; Apache-2.0 for Drizzle ORM and TypeScript; Unlicense for postgres.js (from <code>package-lock.json</code>)	Server and tooling	<code>app/package-lock.json</code>
Hero illustration	Original, drawn from code; replaced a third-party photo	Showcase backdrop	<code>pages/hero/build-hero.py</code>

Item	Licence	How the project uses it	Where
Screenshots	Real captures of the running system	Showcase and deck	app/docs/screenshots/

How the AGPL-3.0 licence of YOLO11 is handled (from docs/raksha/05-dataset-license-verification.md §2 and ai/README.md):

- AGPL-3.0 requires releasing the complete source of a derivative work. For this student project that is acceptable, because the repository is already public coursework.
- For any commercial use the document says plainly that AGPL is a real constraint: either buy Ultralytics' Enterprise Licence or switch to a permissively licensed model (YOLOX and RT-DETR, both Apache-2.0, are named as candidates, but have not been evaluated).
- The serving path already avoids the AGPL dependency: `serve.py` runs the exported ONNX model on ONNX Runtime, NumPy and Pillow only. Training keeps Ultralytics, where nothing is distributed.
- Every Python requirement is pinned, and CI fails if one is not, so the licensing argument stays reproducible.

8

Costs

In plain language

The live project runs without paying for hosting. The free plans have limits, and the biggest one matters: the free server goes to sleep when nobody uses it, which is fine for a demo but not for real emergencies.

Service	Plan	Limits that matter, from the repository
Render	Free web service	Sleeps after 15 minutes idle and takes about a minute to wake. 750 instance hours a month per workspace; one service awake all month is about 720 to 744, so <code>keep-awake.yaml</code> fits only while it is the only free service. The disk is ephemeral (persistent disks need a paid instance), so hazard photos are kept in the database instead (ADR-0013)
Neon	Free	0.5 GB storage, roughly 750 hazard photos at the 600 KiB cap
Cloudflare	Free	"Neon and Cloudflare stay on their free plans" (deploy/aws/README.md)
GitHub (repository, Actions, Pages, GHCR)	No paid plan is referenced anywhere in the repository; the repository and the image are public	Scheduled runs may be delayed under load, so keep-awake means "usually awake"
Resend	Free	The free sender delivers only to the account's own address; a verified domain (<code>send.roadassistbharat.online</code>) lifts that
UptimeRobot	Not recorded in the repository	
Open data services (OSM, Nominatim, Overpass, OSRM, Open-Meteo, USGS)	No account or key	Fair use: paced to about one request a second and cached. The OSRM server is a demo router
Domain <code>roadassistbharat.online</code>	Registered and managed at Hostinger	The price is not recorded in the repository
AWS (PREPARED , not live)	Free plan for accounts created since 15 July 2025	Up to US\$200 in credits (US\$100 at sign-up and up to US\$100 more), valid for 6 months or until the credits run out; then the account must upgrade or AWS closes it. Estimated running cost about US\$14 a month (t3.micro about US\$10, public IPv4 about US\$3.60, 8 GB disk about US\$0.80)
Fly.io (PREPARED , not live)	Not recorded	

The cost of free, said plainly: a free service that sleeps after 15 minutes and takes about a minute to wake is fine for a demo and not fine for the emergency response times the platform describes. A sleeping free tier must not be mistaken for the architecture ([DEPLOYMENT.md](#), `render.yaml`).

9 Glossary

Term	Meaning
API	Application Programming Interface: the set of web addresses (routes) that the apps call to read or change data
PWA	Progressive Web App: a website that can be installed to the home screen and keep working offline
Service worker	A small script the browser runs in the background; here it keeps the app's files and map tiles available offline
IndexedDB	A database built into the browser; here it holds the offline SOS journal
AES-GCM	A standard encryption method; the offline journal is encrypted with a 256-bit key that cannot be read out of the browser
PostgreSQL	An open-source relational database
PostGIS	An extension that adds maps, coordinates and distance searches to PostgreSQL
ORM	Object-Relational Mapper: a library (here Drizzle) that lets code talk to the database with types
Migration	A versioned script that changes the database structure; applied in order, safely repeatable here
JWT	JSON Web Token: a signed token that proves who is calling; here it lasts 10 minutes
Refresh token	A longer-lived token used to get a new JWT; here it rotates on every use
HttpOnly cookie	A cookie that page scripts cannot read, so an injected script cannot steal it
OTP	One-Time Password: the six-digit sign-in code
CSP	Content Security Policy: a header telling the browser which scripts and resources a page may load
CORS	Cross-Origin Resource Sharing: rules for which other websites may call the API
Rate limit	A cap on how many requests one caller can make in a time window
Hash chain	Each audit row stores a fingerprint of itself and the row before, so editing any old row is detectable
SSE	Server-Sent Events: a one-way live connection from the server to the browser for updates
Docker image / container	An image is a packaged program with everything it needs; a container is that image running
Registry (GHCR)	A store for Docker images; GHCR is GitHub's
CI	Continuous Integration: automatic build and test on every change
Workflow	A GitHub Actions file describing jobs to run
Region	The physical data-centre location (here Singapore)
CDN / proxy	A service in front of the server (here Cloudflare) that handles connections and HTTPS
TLS / HTTPS	Encryption for data travelling over the internet
DNS	The internet's address book, turning a domain name into a server address
Modular monolith	One program split into modules with strict boundaries, instead of many small services

Term	Meaning
Idempotent	Safe to repeat: running it twice has the same effect as once (here the off-grid replay and migrations)
YOLO11	A family of real-time object-detection models from Ultralytics
mAP50	Mean Average Precision at 50% box overlap: a standard accuracy score for detectors (0 to 1)
ONNX / ONNX Runtime	An open file format for trained models, and the engine that runs it without the training software
RDD2022	A public road-damage image dataset from six countries
AGPL-3.0	A "copyleft" licence that requires sharing the source of software built on it, even when offered over a network
SIL OFL	The Open Font License: fonts may be used, shared and bundled freely
R8	Android's code shrinker and optimiser for release builds
Jetpack Compose	Android's modern toolkit for building screens in Kotlin
Feature phone	A basic mobile phone with calls and SMS but no apps
DLT	India's TRAI registration that SMS templates must pass before a provider like MSG91 can send them
Least privilege	Giving each account only the rights it needs; here the app's database role cannot change the schema
Penetration test	A deliberate attempt to break into a system; here self-written, not external

Sources: README.md , DEPLOYMENT.md , ENGINEERING-NOTES.md , render.yaml , fly.toml , app/Dockerfile , app/docker-start.sh , app/package.json , app/package-lock.json , app/docs/measured.json , app/docs/SECURITY.md , app/docs/TESTING.md , app/scripts/check-data-residency.mjs , .github/workflows/*.yaml , mobile/build.gradle.kts , mobile/app/build.gradle.kts , mobile/gradle/wrapper/gradle-wrapper.properties , ai/README.md , ai/requirements.txt , ai/requirements-serve.txt , docs/raksha/05-dataset-license-verification.md , deploy/aws/ , ppt/README.md , design/ . Written 2026-10-05.